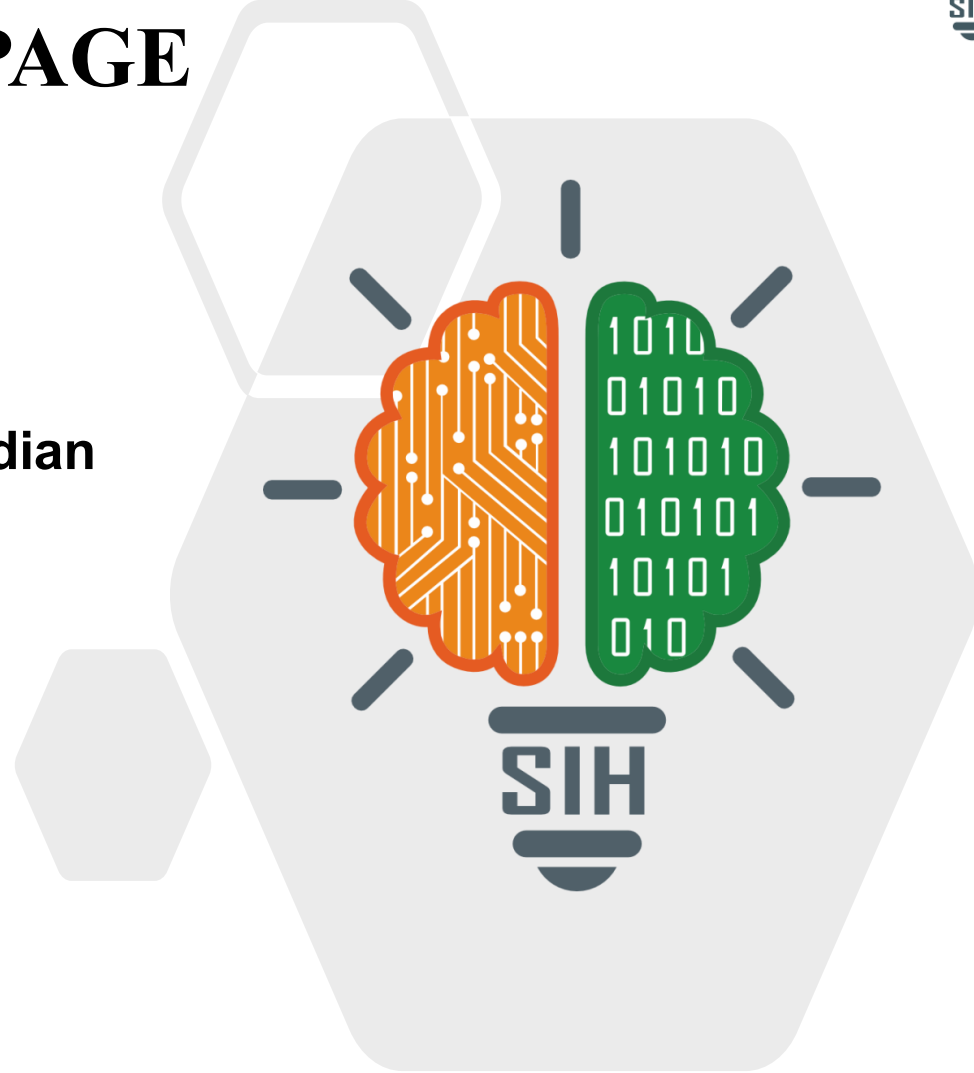




## TITLE PAGE

- Problem Statement ID – SIH26037
- Problem Statement Title – Adaptive Path Planning and Collision Avoidance for Autonomous Vehicles on Unstructured Indian Roads
- Theme – Smart Vehicles
- PS Category – Software
- Team ID – 161638
- Team Name – Ctrl Alt Elite
- Solution – PathSense





## Proposed Solution

**PathSense** is an adaptive autonomous-driving intelligence system for Indian roads where lane markings may be absent and traffic behaviour is irregular.

**Core idea:** perceive every road user, understand its motion, reason about the vehicle's real drivable corridor and keep re-planning a collision-free trajectory.



## Problems It Addresses

**Unmarked Roads:** Lane markings are faded, partial or absent, so lane-following assumptions break.

**Mixed Traffic:** Cars, buses, trucks, two-wheelers, auto-rickshaws, cyclists and pedestrians share one space.

**Informal Merges:** Vehicles cut in from either side without signalling or lane discipline.

**Sudden Direction Changes:** Two-wheelers and autos weave, U-turn or stop abruptly.

**Wrong-Way Movement:** Riders and vehicles travel against the flow on the vehicle's own side.

**Pedestrian / Cyclist Interaction:** People walk along and across the carriageway without crossings.

**Animal Crossings:** Cattle and dogs stand on, or wander into, the road.

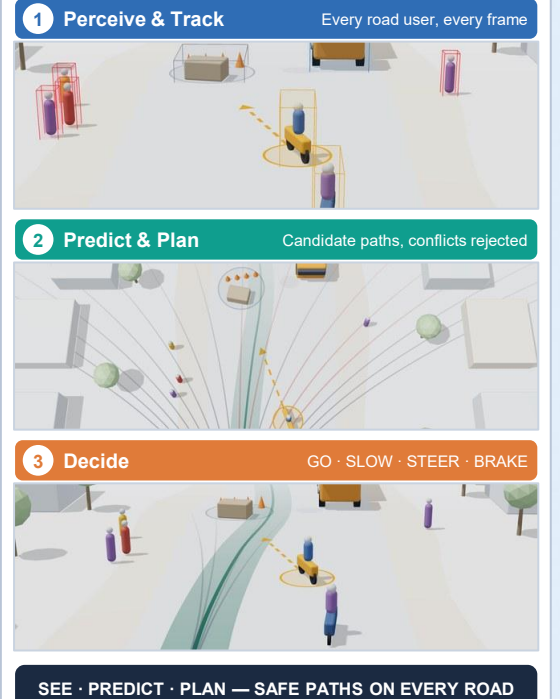
**Temporary Obstacles:** Work zones, parked vehicles, handcarts and debris block the usual path.

## Differentiation & Key Value Proposition

- 01 Lane-Independent Path Planning**  
Plans around the actual drivable corridor instead of depending only on painted lane markings.
- 02 Indian Mixed-Traffic Awareness**  
Designed for cars, buses, trucks, motorcycles, auto-rickshaws, pedestrians, bicycles and animals.
- 03 Behaviour-Aware Collision Avoidance**  
Considers object movement and potential path conflict rather than distance alone.
- 04 Adaptive Replanning**  
Continuously evaluates candidate trajectories as the road scene changes.
- 05 Unstructured-Road Operation**  
Targets roads where conventional lane-based assumptions are unreliable.
- 06 Safety-First Decision Layer**  
Escalates from normal travel to caution, steering, braking and no-safe-path according to path threat.

## How It Works

From the camera view to a safe, explainable path



### Perception

Detects road users & obstacles

### Tracking

Keeps identities & motion over time

### Motion Understanding

Short-term movement & interaction risk

### Ego-Path Understanding

The vehicle's real drivable corridor

### Adaptive Planning

Generates & scores collision-free paths

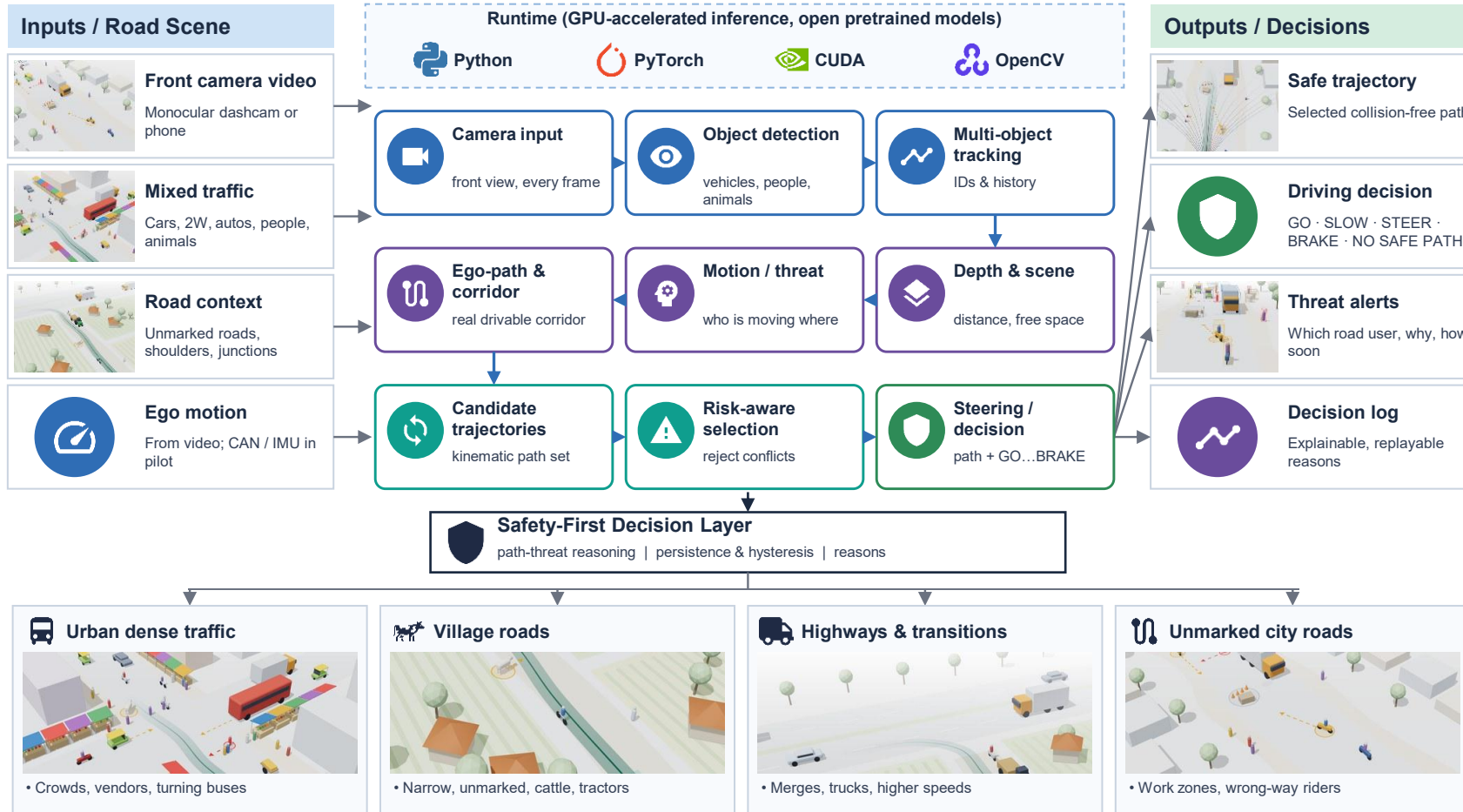
### Decision

GO / SLOW / STEER / BRAKE / NO SAFE PATH

# TECHNICAL APPROACH

PathSense | Adaptive Perception-to-Planning Pipeline

## ARCHITECTURE



## TECH STACK

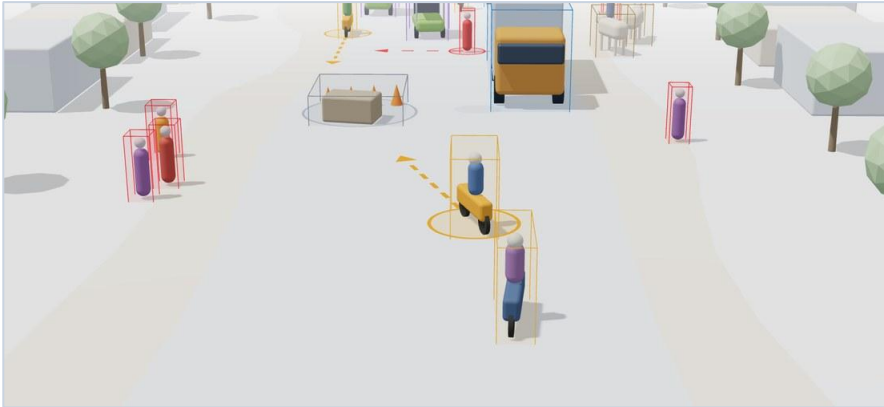
| PERCEPTION                                                                                                              | TRACKING                                                                                         | DEPTH / VISION                                                                                                    |
|-------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| <br>YOLO                             | <br>ByteTrack | <br>Depth Anything             |
| <br>OpenCV<br>Object detection       | Multi-object tracking                                                                            | <br>CLIP<br>Depth & scene cues |
| PLANNING                                                                                                                | WEB / DEMO                                                                                       | RUNTIME                                                                                                           |
| <br>Python                           | <br>Flask     | <br>PyTorch                    |
| <br>NumPy<br>Trajectory & risk logic | <br>HTML      | <br>CUDA<br>GPU inference      |
|                                                                                                                         | <br>CSS       | <br>JS                         |
|                                                                                                                         | <br>CSS       | Review dashboard                                                                                                  |

## METHODOLOGY



## Concept Views

### Perception View



Road users detected as 3D boxes; predicted motion as dashed arrows

### Planning View



Candidate paths fanned out; conflicting paths (red) rejected; safe path (teal) kept

## Before vs After



## Current Development

**Working prototype:** camera-based perception, tracking, motion understanding and adaptive path planning

**Code repository:**

[github.com/MalyalaKarthik66/pathsense-prototype](https://github.com/MalyalaKarthik66/pathsense-prototype)

Live demonstration shown separately during evaluation.

### What the prototype demonstrates

- Detection & tracking of mixed Indian road users
- Path-threat reasoning: oncoming vs. entering the path
- Adaptive candidate-path planning around obstacles
- Explainable decisions with replay

Visuals on this slide are original concept renders, not application screenshots.

## RISKS → MITIGATION

- **Unstructured road geometry** → Drivable-corridor estimation and path-aware planning.
- **Unpredictable road-user behaviour** → Temporal tracking and short-term motion reasoning.
- **Perception uncertainty** → Confidence-aware decisions and future sensor fusion.
- **Real-world validation complexity** → Scenario library + simulation + controlled field testing.

## FEASIBILITY

- **Technology:** Established computer-vision and motion-planning building blocks; camera-based perception to start.
- **Architecture:** Modular perception → prediction → planning; radar / LiDAR and vehicle-state inputs added progressively; scenario-based simulation supports development.

## VIABILITY

- **Relevance:** Mixed, irregular traffic on urban roads, village roads, highways and dense traffic.
- **Deployment:** Modular architecture supports incremental deployment through controlled pilots.
- **Future fit:** Can integrate with future intelligent-vehicle platforms.

## PROTOTYPE

- **Camera-based proof of concept** for perception, tracking and adaptive path planning.

## PILOT

- **Multi-scenario validation** with richer sensing, simulation and controlled road testing.

## PRODUCTION

- **Vehicle-grade multi-sensor integration**, closed-loop validation and automotive safety engineering.

## Potential Collaborators / Partners



**Automotive OEMs & Tier-1s**  
ADAS / AV integration



**Research institutes**  
IITs, IISc, AV labs



**Simulation partners**  
e.g. MathWorks toolchain



**Road-safety bodies**  
MoRTH, state transport



**Fleet operators**  
Buses, taxis, logistics



**Mapping & data**  
OpenStreetMap, IDD



## POTENTIAL IMPACTS

### Target Audience

**Drivers / Passengers:** Improved handling of unpredictable road interactions

**Urban Mobility:** More adaptive navigation in dense mixed traffic

**Rural Mobility:** Better operation on roads with limited lane structure

**Road Safety:** Earlier recognition of path conflicts and adaptive response

**Future Autonomous Vehicles:** A foundation for Indian-road-aware navigation



## BENEFITS

### SAFETY

- Path-aware collision avoidance
- Earlier recognition of path conflicts

### ADAPTABILITY

- Handles changing road and traffic conditions
- Works where lane markings are unreliable

### SCALABILITY

- Can evolve from prototype → pilot → production
- Modular: new sensors and platforms plug in

### INDIAN-ROAD RELEVANCE

- Designed around mixed and irregular road behaviour
- Safer shared roads for pedestrians and two-wheelers

## PoC

## PROOF OF CONCEPT

**1**

### PROBLEM

Unstructured Indian roads create difficult path-planning conditions

**2**

### OBJECTIVE

Safe adaptive trajectories without relying solely on lane markings

**3**

### PROTOTYPE

Camera-based perception, tracking, motion understanding and adaptive planning

**4**

### VALIDATION

Scenario-based evaluation across mixed-traffic road situations

**5**

### SUCCESS CRITERIA

Safe path selection, appropriate risk response, robust behaviour across roads

**6**

### NEXT STEPS

Sensor fusion, closed-loop simulation, broader field trials, vehicle integration

**Resources:** computer vision & motion planning, road-scene video, simulation tools, compute hardware

# BUSINESS MODEL CANVAS

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div><div>Key Partnerships</div><div></div><div><ul style="list-style-type: none"><li>• <b>OEMs:</b> Automotive OEMs &amp; Tier-1 ADAS suppliers</li><li>• <b>ACADEMIA:</b> IITs, IISc &amp; autonomy research labs</li><li>• <b>SIMULATION:</b> MathWorks toolchain (RoadRunner, Automated Driving Toolbox)</li><li>• <b>DATA:</b> Indian driving datasets (IDD), OpenStreetMap</li><li>• <b>GOVT:</b> MoRTH, state transport &amp; road-safety bodies</li><li>• <b>FLEETS:</b> Logistics, taxi &amp; public-transport operators</li></ul></div></div> | <div><div>Key Activities</div><div></div><div><ul style="list-style-type: none"><li>• <b>PERCEPTION R&amp;D:</b> Detection, tracking &amp; depth</li><li>• <b>PLANNING:</b> Adaptive trajectory &amp; risk logic</li><li>• <b>VALIDATION:</b> Scenario library &amp; simulation</li><li>• <b>INTEGRATION:</b> Vehicle / ADAS interfaces</li></ul></div></div> <div><div>Key Resources</div><div></div><div><ul style="list-style-type: none"><li>• <b>SOFTWARE STACK:</b> Perception-to-planning pipeline</li><li>• <b>DATA:</b> Indian-road scenario library</li><li>• <b>TEAM:</b> CV, robotics &amp; software engineers</li><li>• <b>TOOLS:</b> Simulation &amp; compute infrastructure</li></ul></div></div> | <div><div>Value Propositions</div><div></div><div><ul style="list-style-type: none"><li>• <b>OEMs:</b> An Indian-road-aware planning layer</li><li>• <b>FLEETS:</b> Fewer conflicts in mixed traffic</li><li>• <b>ROAD USERS:</b> Earlier recognition of path conflicts</li><li>• <b>RESEARCHERS:</b> Scenario-based validation workflow</li><li>• <b>EXPLAINABLE:</b> Every decision comes with its reason</li><li>• <b>MODULAR:</b> Camera-first, sensor-ready architecture</li></ul></div></div> | <div><div>Customer Relationships</div><div></div><div><ul style="list-style-type: none"><li>• <b>CO-DEVELOPMENT:</b> Pilot programmes with OEMs</li><li>• <b>SUPPORT:</b> Integration &amp; tuning</li><li>• <b>UPDATES:</b> Continuous model &amp; scenario updates</li><li>• <b>TRANSPARENCY:</b> Decision logs for audit</li></ul></div></div> <div><div>Channels</div><div></div><div><ul style="list-style-type: none"><li>• <b>DIRECT B2B:</b> OEM &amp; Tier-1 partnerships</li><li>• <b>PILOTS:</b> Fleet &amp; campus deployments</li><li>• <b>RESEARCH:</b> Academic collaborations</li><li>• <b>SDK / API:</b> Licensable software modules</li></ul></div></div> | <div><div>Customer Segments</div><div></div><div><ul style="list-style-type: none"><li>• <b>PRIMARY:</b> Automotive OEMs &amp; ADAS suppliers</li><li>• <b>SECONDARY:</b> Fleet &amp; logistics operators</li><li>• <b>PUBLIC:</b> Transport &amp; road-safety agencies</li><li>• <b>RESEARCH:</b> AV labs &amp; testing bodies</li><li>• <b>SCALE-UP:</b> City → state → national deployments</li></ul></div></div> |
| <div><div>Cost Structure</div><div></div><div><ul style="list-style-type: none"><li>• <b>R&amp;D</b> – perception &amp; planning engineering</li><li>• <b>Data</b> – collection, annotation &amp; scenario building</li><li>• <b>Compute</b> – training &amp; simulation infrastructure</li><li>• <b>Validation</b> – field testing &amp; safety engineering</li><li>• <b>Operations</b> – integration, support &amp; updates</li></ul></div></div>                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <div><div>Revenue Streams</div><div></div><div><ul style="list-style-type: none"><li>• <b>Software licensing</b> – per-vehicle / per-platform licences for OEMs</li><li>• <b>Pilot contracts</b> – fleet &amp; campus deployments</li><li>• <b>Scenario-library access</b> – validation datasets &amp; tools</li><li>• <b>Integration services</b> – customisation &amp; calibration</li><li>• <b>Support &amp; updates</b> – annual maintenance subscriptions</li></ul></div></div>              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

## SUPPORTING MATERIAL

## SUPPORTING RESEARCH PAPERS

## AUTONOMOUS DRIVING &amp; PATH PLANNING

"A Survey of Motion Planning and Control Techniques for Self-Driving Urban Vehicles" Paden et al. — IEEE T-IV, 2016  
[arxiv.org/abs/1604.07446](https://arxiv.org/abs/1604.07446)

## DEPTH ESTIMATION

"Depth Anything V2" Yang et al. — NeurIPS, 2024  
[arxiv.org/abs/2406.09414](https://arxiv.org/abs/2406.09414)

## OBJECT DETECTION &amp; TRACKING

"ByteTrack: Multi-Object Tracking by Associating Every Detection Box" Zhang et al. — ECCV, 2022  
[arxiv.org/abs/2110.06864](https://arxiv.org/abs/2110.06864)

## MOTION PREDICTION

"Social LSTM: Human Trajectory Prediction in Crowded Spaces" Alahi et al. — CVPR, 2016  
[openaccess.thecvf.com](https://openaccess.thecvf.com)

## INDIAN ROAD DATA

## IDD – Indian Driving Dataset

[idd.insaan.iiit.ac.in](http://idd.insaan.iiit.ac.in)

## MoRTH – Road Accidents in India

[morth.nic.in](http://morth.nic.in)

## OpenStreetMap

[openstreetmap.org](https://openstreetmap.org)

## Wikimedia Commons

[commons.wikimedia.org](https://commons.wikimedia.org)

## SIH &amp; CONTEXT

## SIH 2026 – PS SIH26037

[sih.gov.in](http://sih.gov.in)

## MathWorks – RoadRunner &amp; ADT

[mathworks.com](https://mathworks.com)

Problem focus: unmarked roads, mixed traffic, informal merges, obstacles.

## TECHNICAL DOCS

## Ultralytics YOLO

[docs.ultralytics.com](https://docs.ultralytics.com)

## PyTorch

[pytorch.org/docs](https://pytorch.org/docs)

## OpenCV

[docs.opencv.org](https://docs.opencv.org)

## Flask

[flask.palletsprojects.com](https://flask.palletsprojects.com)

## NATIONAL ALIGNMENT

## Road Safety

## MoRTH road-safety goals

Earlier recognition of path conflicts on Indian roads

## IndiaAI Mission

## AI for Indian conditions

Perception and planning built for Indian traffic

## Atmanirbhar Bharat

## Self-reliant technology

Home-grown autonomous-driving intelligence

## Make in India

## Indigenous automotive tech

ADAS software developed for Indian vehicles

## Smart Cities Mission

## Intelligent urban mobility

Adaptive navigation in dense city traffic

## Digital India

## Software-first innovation

Open, modular and explainable mobility software

Sources: IEEE T-IV · ECCV · NeurIPS · CVPR · IDD (IIIT-H) · MoRTH · OpenStreetMap · SIH 2026 · Ultralytics · PyTorch · OpenCV